



QUALITY SYSTEM

FORM

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TITLE:

LABORATORY OOS INVESTIGATION FORM

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OOS Number: OOS- 262017

Phase I

Section A – Test Details

Initiator:	Cuong Du (Written by Qiyue C)	Test Date:	24-Aug-2026	Date Initiated:	31-Aug-2026
Date of Incident:	31-Aug-2026	Test Name:	Celsis Sterility Test	Sample ID:	ETX-260821-0259
Name of Analyst who Performed the Test:	Prepping Analyst: Elysse Nioupin (EN)	Sample / Active Name	Tesamorelin 12 mg per Vial	Lot #:	16590773
	Processing Analyst: Elizabeth Sanchez (ES)				
	Aliquoting Analyst: America Alanis (ALA)	Dosage Form	Lyophilized		
Description of Incident:	On 31 Aug 2026, sample ETX-260821-0259 was found to be positive for growth after Celsis Sterility Testing		☒ Client Care notified of OOS ☒ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS		
	SOP / Test Method # MICRO-SOP-44	Effective Date: 03-Jan-2025	SOP / Test Method Rev: 01	Limits/Specification: Pass/Fail	
Manager notified	Manager/designee name: Kathan Parikh	Date: 31-Aug-2026	Quality Assurance Department notified:	Yes ☒ No ☐	

Section B – Investigation and Root Cause Analysis

Personnel Discrepancy		Comments
Analyst interviewed?	☒ Yes ☐ No ☐ N/A	Yes, analysts Elysse Nioupin, Elizabeth Sanchez, America Alanis & Cuong Du were comprehensively interviewed.
Correct samples analyzed?	☒ Yes ☐ No ☐ N/A	Yes, Sample ID ETX-260821-0259
Correct SOP / Test Method used/followed?	☒ Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-44
Correct sample collection/technique?	☒ Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-44
Are the Test Result Records Attached & Calculations Verified and Correct?	☒ Yes ☐ No ☐ N/A	Yes, Information is available in EagleTrax under ETX-260821-0259
Is the analyst qualified to perform the test?	☒ Yes ☐ No ☐ N/A	Yes, all analysts are trained and qualified by the quality department to perform the test
Is formula worksheet provided and verified?	☐ Yes ☐ No ☒ N/A	Not Applicable
Does the processing method/method set conform to the required specification?	☐ Yes ☐ No ☒ N/A	Not Applicable
Was the sample stored appropriately?	☒ Yes ☐ No ☐ N/A	Yes, the sample was stored refrigerated as per client's instructions
Material Discrepancy		Lot Number
Is the integrity of the sample(s) questionable?	☐ Yes ☒ No ☐ N/A	
Was the sample transported appropriately?	☒ Yes ☐ No ☐ N/A	
Are the other test results from this run under investigation?	☐ Yes ☒ No ☐ N/A	



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Appropriate/correct glassware used?	☐ Yes ☐ No ☒ N/A		
Are reagents stored in proper condition and not expired?	☒ Yes ☐ No ☐ N/A	Materials and Reagents TSB: 07102026-1 FTM: 06232026-5 Tween 80: 0000606420 Celsis Reagents & Kits: Please refer to attached packet TSA / Surface Plates: See EM logs Monthly Cleaning: H2O2 strips: 250409 IPA: 01600581	Materials and Reagents TSB: 08 Oct 2026 FTM: 21 Sep 2026 Tween 80: 01 Sep 2027 Celsis Reagents & Kits: Please refer to attached packet TSA / Surface Plates: See EM logs Monthly Cleaning: H2O2 strips: Apr 2027 IPA: Sep 2027
Standard Used	Celsis ATP Positive Control	022601-1483	Jan 2027
Solvent Used	Not Applicable	Not Applicable	Not Applicable
Is the standard the source of the root cause?	☐ Yes ☐ No ☒ N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐ Yes ☐ No ☒ N/A		
Equipment Discrepancy		Equipment ID	Calibration Due Date
Was the instrument set up correctly?	☒ Yes ☐ No ☐ N/A	E002222	Sep 2026
Is the clean room facility certified?	☒ Yes ☐ No ☐ N/A	E001736 (CR114) BSC E001316 E001736 (CR114) BSC E001708	Dec 2026 / Jun 2027 Dec 2026 / Jun 2027
System Performance Check (if applicable)?	☒ Yes ☐ No ☐ N/A	E002222	Sep 2026
System Suitability RSD < 2% (PDA)?	☐ Yes ☐ No ☒ N/A	Not Applicable	Not Applicable
System Suitability RSD < 10% (QDA, CAD)?	☐ Yes ☐ No ☒ N/A	Not Applicable	Not Applicable
Do the environmental monitoring results conform to required standards?	☒ Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☒ Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☒ Yes ☐ No ☐ N/A	Incubator E001356 (Sensor E001450) Incubator E001357 (Sensor E001449) Incubator E001034 (Sensor E001501) Incubator E001031 (Sensor E001505)	Jan 2027 / Feb 2027 Jan 2027 / Feb 2027 Aug 2027 / Feb 2027 Aug 2027 / Feb 2027

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Are chromatogram/spectra/raw data correct?				☐ Yes	☐ No	☒ N/A	Not Applicable	Not Applicable
Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)								
☐ N/A		☐ Re-vialing of original working sample solution						
☐ Re-injection of sample vial		☐ Re-dilution of a working sample solution from the original stock						
☐ Re-injection of original standard vial		☐ Re-dilution of a working standard solution from the original stock standard						
☐ Re-vialing of original working standard solution		☐ Other: N/A CD 31Aug2026						
Section D – Phase I Summary								
Phase I Summary: All analysts involved in the preparation, processing, aliquoting, and reading of the samples, Elysse Nioupin, Elizabeth Sanchez, America Alanis, and Cuong Du, were interviewed comprehensively. Their responses are documented throughout this report. This investigation was performed as per MICRO-SOP-53 Sterility Test Out-of-Specification (OOS) Investigation Procedure. Upon arrival on 21 Aug 2026, the sample was stored refrigerated in accordance with the client instructions. Both sample prepping analyst Elysse Nioupin and processing analyst Elizabeth Sanchez inspected the sample during preparation and processing. No leaks or visible turbidity were observed, and no evidence of compromised container integrity was identified. All reagents and supplies mentioned in the material section above were stored according to supplier recommendations, and their integrity was visually verified before utilization. Moreover, all reagents and supplies had valid expiration dates. The functionality of all equipment was confirmed by reviewing data generated by our comprehensive in-house continuous monitoring system. The cleanroom suite used for processing procedures (CR114) comprises three interconnected sections: the innermost ISO 7 cleanroom (114B), which connects to the middle ISO 7 buffer room (114A), and then to the outermost ISO 8 anteroom (114). A positive air pressure system is maintained throughout the suite to ensure controlled, unidirectional airflow from 114B through 114A and into 114. The innermost ISO 7 cleanroom 114B and its ISO 5 BSC E001316 were thoroughly cleaned and prepared before initiating testing by analyst Elizabeth Sanchez as per MICRO-SOP-2 (Environmental Monitoring of the Clean Room Facility) and MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). For Celsis aliquoting, intermediate ISO 7 buffer room 114A and its ISO 5 BSC E001798 were thoroughly cleaned and prepared by analyst America Alanis before initiating testing as per MICRO-SOP-44 and MICRO-SOP-9. Both BSCs, E001316 in 114B and E001798 in 114A, were certified and approved by the Engineering and Quality Assurance teams prior to use.								

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Phase I Summary (continued):

On 24 Aug 2026, the sample vials for ETX-260821-0259 were received from the Sample Submissions team and brought into the Sterile Microbiology lab. Upon arrival, each vial was sprayed with an acidified bleach disinfectant, placed into pre-disinfected bins, and allowed a 10-minute contact time. Secondary disinfection was performed in the ISO 8 anteroom (114), where the vials were again treated with acidified bleach with a 10-minute contact time before transfer into cleanroom 114B. Inside 114B, processing analyst Elizabeth Sanchez performed a final disinfection step with a 10-minute contact time. Once fully disinfected, the vials were transferred into the ISO 5 BSC E001316, placed on the disinfected work surface, aseptically opened, and tested in accordance with MICRO-SOP-44. Direct inoculation was performed by adding a total of 84 mL of sample across 14 vials (6 mL per vial), using nine 300 mL FTM jars and nine 300 mL TSB jars, in accordance with suitability method record: ETX-251218-0432. Tween 80 (Lot# 0000606420, Exp: 01Sep2027) was incorporated into the test procedure per suitability requirements. Accordingly, a matching negative control with the same modifications was prepared and handled identically to actual samples. Following testing, the media bottles were transferred into incubators E001356 and E001357 to initiate incubation.

Upon completion of incubation on 31 Aug 2026, the media bottles for ETX-260821-0259 were disinfected and transferred to the middle ISO 7 buffer room (114A) for the aliquoting step per MICRO-SOP-44 (Celsis Sterility Testing). In cleanroom 114A, the media bottles were disinfected one more time before transfer into the ISO 5 BSC E001798 located in cleanroom 114A. In ISO 5 BSC E001798, the sample was aliquoted into assay cuvettes by analyst America Alanis. Following aliquoting, Celsis Sterility Reading was performed in accordance with MICRO-SOP-44 by analyst Cuong Du on instrument Advance 2 (E002222).

Following the reading, sample ETX-260821-0259 was found to yield positive readings in two of the 300 mL TSB media jars (Bottle 1 and Bottle 6). Bottle 1 yielded > 9,999,999 RLU (instrument overload) on both the initial read and confirmation re-read. Bottle 6 yielded 1,275 RLU (duplicate reading tubes 1,301 RLU and 1,249 RLU, %CV 2%) on the initial read and 1,386 RLU (duplicate reading tubes 1,481 RLU and 1,291 RLU, %CV 7%) upon confirmation re-read, both exceeding the method cutoffs of 1,197.0 RLU and 1,165.5 RLU respectively, where the corresponding TSB negative control yielded 399 RLU and 389 RLU. All other TSB bottles (Bottles 2, 3, 4, 5, 7, 8, 9) and all nine FTM sample bottles in the same batch tested negative (FTM cutoff 5,416.5 RLU, FTM negative control 1,806 RLU). The %CV from duplicate reading tubes for the positive TSB bottles was well within specification (< 30%). Additionally, all Daily Controls, including Instrument Blank (8 RLU), Reagent Blank (77 RLU, %CV 1%), and ATP Positive Control (103,864 RLU, %CV 3%), met defined acceptance criteria with %CV below 30%.

Following the OOS result, the positive TSB bottles for ETX-260821-0259 were submitted for Differential Staining and Microbial Identification under ETX-260831-0608. Microbial growth was successfully recovered by subculture on solid growth media, and differential staining visualized Gram-positive rods. Microbial identification confirmed the isolate as *Terribacillus goriensis*, as presented in Table 1.

The culture media utilized were within their expiry period (TSB lot 07102026-1, Exp: 08Oct2026; FTM lot 06232026-5, Exp: 21Sep2026). The negative culture media bottles for the direct inoculation method for the original culture were handled, processed, and incubated in a manner identical to that of actual samples. No microbial growth was observed in the corresponding negative control.

Monthly cleaning and disinfection of the outermost ISO 8 Anteroom, the middle ISO 7 Buffer room, the innermost ISO 7 cleanroom, and its containing ISO 5 Biosafety Cabinets for CR114 was performed on 16 Aug 2026, as per MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). It was documented that all H2O2 indicators passed. This confirms the efficient monthly cleaning of all three parts of Cleanroom 114 prior to testing.

Attached Tables 2 and 3 present the environmental monitoring results for the duration of testing. The environmental monitoring (EM) plates were incubated for no less than 48 hours at 30-35°C and no less than an additional five days at 20-25°C as per MICRO-SOP-2 (Environmental Monitoring of the Cleanroom Facility). Table 2 pertains to Environmental Monitoring performed during Celsis Sterility Processing, and Table 3 pertains to Environmental Monitoring performed during Celsis Sterility Aliquoting.

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A review of the Environmental Monitoring (EM) results for the relevant Celsis Sterility processing period showed no microbial growth on personnel monitoring plates, settling plates, or ISO 5 BSC E001316 surface sampling plates on the date of testing (24Aug26), the preceding sampling date (21Aug26), or the subsequent sampling date (25Aug26). Weekly active air and surface sampling of the cleanroom suite associated with processing was performed on 25Aug26. No microbial recovery was observed from weekly surface sampling of the cleanroom suite. Active air sampling showed recovery of 4 CFU (ETX-260901-0112) in the ISO 8 area, identified as 3 Gram (+) cocci and 1 Hyphae, with zero recovery observed in the ISO 7 buffer or cleanrooms.

After reviewing the Environmental Monitoring (EM) results for the relevant Celsis Sterility aliquoting period, no microbial growth was observed on personnel monitoring plates, ISO 5 BSC E001798 surface sampling plates, or settling plates on the date of aliquoting (31Aug26), the preceding sampling date (28Aug26), or the subsequent sampling date (01Sep26). Weekly active air sampling of the cleanroom suite performed on 04Sep26 showed recovery of 2 CFU (ETX-260914-0487) in the ISO 8 area, identified as *Corynebacterium ureicelerivorans* and *Mycobacterium grossiae*, with zero recovery in the ISO 7 buffer or cleanrooms, and weekly surface sampling of the cleanroom suite showed no growth.

Evaluation of Environmental Monitoring Results: It is important to note that, for both processing and aliquoting activities, the samples are processed within validated ISO 5 BSCs (E001316 and E001798 respectively). Although microbial recovery was observed from weekly active air monitoring in the outermost ISO 8 room 114 background environment in cleanroom suite (CR114), these recoveries were not obtained from the ISO 5 processing environment or the ISO 7 cleanroom environment.

The absence of microbial recovery from both the processing and the aliquoting analyst glove plates, ISO 5 BSC surface monitoring plates, and ISO 5 settling plates indicates that no identifiable contamination transfer pathway existed between the monitored background cleanroom environment and the ISO 5 critical processing areas. Samples were also prepared and transferred into disinfected, lidded containers and transported on carts within the controlled cleanroom environment. Therefore, while the active air results demonstrate microbial recovery in the surrounding ISO 8 area, there is no evidence that the recovery originated from, or affected, the ISO 5 BSC environment where the test sample was handled and processed.

All analysts confirmed full compliance with cleaning procedures as outlined in MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology) and MICRO-SOP-44 (Celsis Sterility Testing). A review of the available data confirms that the cleanroom and equipment conditions remained within acceptable parameters. No deviations or obvious signs of laboratory error during processing and aliquoting were noted from analysts.

No other sample processed by analyst Elizabeth Sanchez on 24 Aug 2026 and aliquoted by analyst America Alanis on 31 Aug 2026, failed Celsis Sterility testing that day. To assess the potential for sample-to-sample contamination contributing to the positive results, a comprehensive review was conducted of all samples processed in the same batch. All other samples processed by the same analyst on the day of testing were found to test negative for microbial growth. These findings suggest that cross-contamination between samples is unlikely.

Analyzing a 6-month sample history for Revive Rx Pharmacy - PO Required (E00927) indicates that this specific analyte "Tesamorelin 12 mg per Vial" (Lot: 16590773) has had no prior failures using the Celsis Sterility testing during this period, and review of lot history confirms no other submissions of this lot failed testing.

Based on the laboratory investigation, no assignable laboratory cause was identified involving the analyst, instrument, reagents, supplies, or monitored laboratory environment. The initial Celsis OOS result therefore remains valid in accordance with the applicable laboratory OOS procedure.



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Has root cause been identified as a possible laboratory error?*Note: Root cause must be conclusive, verifiable, and well-documented.*☐ Yes☐ If retesting is **required**, complete Phase II of the OOS (initiate Form 3.100.019.F02).☒ No, the OOS result is valid.**Reviewer comment(s):**

Not Applicable

Prepared by (Print):

Qiyue Chen

Signature/Date:

SIGN

Lab Manager/Designee (Print):

Robin Seymour

Signature/Date:

SIGN

Section E – Quality Disposition**Can investigation be closed?**☐ Yes☐ Original result is valid.☐ No, initiate Form 3.100.019.F02.**Approved by QA Representative (Print):**

Signature/Date:

SIGN